



Activities & Assessments

Programming and Artificial Intelligence

CHAPTER 4 | PART ONE | FIRST SEMESTER

Unit 4: Web Design and Media

4 lessons | Performance tasks | Home assignments | Weekly assessments

CHAPTER 4

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Unit 4: Web Design and Media

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HOW THIS WORKBOOK IS ORGANISED

1

Classroom Performances *(continuous formative assessment)*

Varied activities the teacher uses during the lesson to boost student interaction, check progress directly, and identify strengths and areas needing improvement.

2

Home Assignments *(mastery and self-application)*

Guided activities and exercises that help students learn independently, check their understanding, and apply what they have learned in new contexts.

3

Weekly Assessments *(performance measurement and follow-up)*

Assessment models carried out at the end of each week to track progress and measure the learning outcomes achieved during that week.

Part I of each period contains the classroom *Performance Tasks*; **Part II** contains the *Home Assignments*. Period 3 of every lesson holds the *Weekly Assessments* in three models (A, B and C). Each question has dotted lines for your written answer.

UNIT 4 | WEB DESIGN AND MEDIA

LESSON

1

Types and Characteristics of Media

PERIOD 1

Week:

Date:/...../.....

Part I | Performance Tasks

Essay Questions:

1. Compare the characteristics of television and radio in terms of the type of information conveyed and the mode of communication.

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2. Explain the main strengths and constraints of websites compared with traditional media.

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Multiple-Choice Questions: (choose the correct answer)

1. Which of the following media is characterized by conveying audio only, and is usually one-way, with the ability to listen to it while on the move?
 - A. Television.
 - B. Radio.
 - C. Newspaper.
 - D. Website.
2. What is one constraint of websites?
 - A. Their information cannot be updated at any time.
 - B. They lack two-way interaction.
 - C. Most of their functions need an Internet connection.
 - D. They are limited to displaying text only, without multimedia.

Part II | Home Assignments

Essay Questions:

1. Explain the purpose and the suitable medium when wanting to inform a large number of people at once using video.

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2. Why do organizations, in practice, use several media together (such as directing viewers from a TV ad to a website) instead of just one?

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Multiple-Choice Questions: (choose the correct answer)

1. What is the most suitable medium mentioned for delivering direct, personalized promotional-campaign information to specific people?
 - A. Television.
 - B. Website.
 - C. Email or direct messages.
 - D. Radio.
2. What is meant by the term “two-way communication” in digital media?
 - A. Communication that goes from sender to receiver only, without any reply.
 - B. The ability for users to respond and participate through forms and comments.
 - C. Reading printed content without any Internet connection.
 - D. Listening to an audio broadcast while doing other tasks.

PERIOD 2

Week: Date:/...../.....

Part I | Performance Tasks**Essay Questions:**

1. How does a newspaper differ from a website in terms of the media types available and the ability to communicate and update?
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.....
2. Name a suitable medium for achieving wide and fast spread of information, stating the reason for your choice.
.....
.....

Multiple-Choice Questions: (choose the correct answer)

1. Which of the following media remains physically present as a printed medium and can be read without an Internet connection?
 - A. Television.
 - B. Social media.
 - C. Newspaper.
 - D. Website.
2. Which of the following represents the most suitable purpose for using social media?
 - A. Presenting detailed, organized information across multiple pages and sections.
 - B. Providing printed information with no connection.
 - C. Spreading by word of mouth, and posting and sharing information quickly.
 - D. Delivering individual, direct information to one specific person only.

Part II | Home Assignments**Essay Questions:**

1. Explain how a website can combine diverse information and support two-way interaction.
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2. Name two possible drawbacks or constraints of using a website as the main channel for a service organization addressing families who may not have home Internet access.
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.....

Multiple-Choice Questions: (choose the correct answer)

1. What distinguishes social media from traditional media such as television and radio?
 - A. It is limited to one direction from sender to receiver.
 - B. Its ability to spread information quickly and to let users post, share, and interact.
 - C. Being a printed medium that remains physically present without needing electricity.
 - D. Conveying audio only, without any images or text.
2. When choosing the suitable medium to convey information, what does the choice mainly depend on?
 - A. Always preferring printed media in every case.
 - B. The purpose, the target audience, and the available resources.
 - C. Avoiding Internet use completely.
 - D. Relying on a single medium for all types of audiences.

PERIOD 3**Weekly Assessments**

Week:

Date:/...../.....

MODEL A**Essay Questions:**

1. Compare in detail the characteristics of television and newspapers in terms of the type of information conveyed, the mode of communication, and reliance on the Internet.
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.....
2. Explain the five main strengths of websites.
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3. Show how different media integrate in practice, giving a practical example that combines television, a website, and social media.
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.....

4. Discuss the three constraints of websites, explaining the effect of each constraint on the user experience.

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Multiple-Choice Questions: (choose the correct answer)

- Which medium combines video and audio, and is usually one-way from sender to receiver, to reach a large audience?

A. Newspaper.	B. Television.
C. Radio.	D. Email.
- Which of the following is a strength of websites?

A. Limited to one permanent direction and does not accept comments.	B. Inevitably needs an Internet connection for all its functions, without exception.
C. Its information can be updated at any time, and it can interact with users.	D. Images or video cannot be shown on it.
- What is the most suitable medium if the purpose is to make a large amount of detailed information available, organized across multiple pages and sections?

A. Radio.	B. Website.
C. Printed newspaper.	D. Direct text messages.
- The medium that allows users to post and share information quickly is:

A. Newspaper.	B. Television.
C. Social media.	D. Radio.

MODEL B

Essay Questions:

- Explain the characteristics of radio, showing its advantages for listening while on the move and its one-way communication constraints.

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- Explain how websites can provide two-way interaction, showing the tools used for that.

.....

.....
- State the suitable purpose for each of the following media: television, website, email, social media.

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.....

4. Discuss the challenges facing organizations trying to reach an audience without home Internet, and the available alternative media solutions.

Multiple-Choice Questions: (choose the correct answer)

1. A medium that relies on audio only and can be listened to while performing other tasks, depending on coverage, is:
 - A. Television.
 - B. Radio.
 - C. Website.
 - D. Newspaper.
2. A technology or medium that allows individual, direct delivery to specific people is:
 - A. Television.
 - B. Social media.
 - C. Email or direct messages.
 - D. Daily newspaper.
3. Among the constraints of websites explained in the lesson:
 - A. It is impossible to update their content after creation.
 - B. The user often needs to open the site to access its content, unless notifications are available.
 - C. Images and text cannot be combined together.
 - D. It is entirely a printed paper medium.
4. Which of the following statements accurately expresses the main idea of choosing media?
 - A. There is one ideal medium required for all purposes at all times.
 - B. There is no single medium that is best for everything; the choice should be based on purpose and audience, combining media where useful.
 - C. Television alone should always be used to avoid the cost of the Internet.
 - D. Social media completely replaces any website or other medium.

MODEL C

Essay Questions:

1. Discuss the role of printed newspapers and their characteristics compared with modern digital media in terms of communication and physical presence.
2. Explain how the suitable medium is chosen based on an analysis of the audience, the purpose, and the target message.
3. Discuss the importance of combining different media in media campaigns and effectively directing the audience between them.

4. Explain the meaning of the terms “one-way communication” and “two-way communication”, with an example of each.

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Multiple-Choice Questions: (choose the correct answer)

1. A printed medium that remains physically present, can be read without an Internet connection, and consists of text and images, is:

A. Television.	B. Newspaper.
C. Website.	D. Radio.
2. Which of the following media is most suitable for achieving wide spread through word of mouth and user sharing?

A. Radio.	B. Printed newspaper.
C. Social media.	D. Individual email.
3. One of the basic characteristics of a website is that it is:

A. Basically one-way, but two-way interaction is possible through forms and comments.	B. Conveys audio only and does not support text or images.
C. Can never be updated after its first publication.	D. Impossible to access through mobile-phone browsers.
4. The basic purpose of using TV ads or social-media ads is to:

A. Present precise, encyclopedic details across multiple pages.	B. Inform a large number of people at once, with wide reach.
C. Send a secret message to one person only.	D. Read long articles without the Internet.

UNIT 4 | WEB DESIGN AND MEDIA

LESSON

2

Information Design and User Experience for Websites

PERIOD 1

Week:

Date:/...../.....

Part I | Performance Tasks

Essay Questions:

1. Explain what is meant by the concept of “persona”, and show how it helps the team in the design process without relying on personal assumptions.
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.....
2. Explain the basic purpose of using a “wireframe” before moving to visual design, and why deciding it first is a necessary step for reaching agreement among stakeholders.
.....
.....

Multiple-Choice Questions: (choose the correct answer)

1. Which of the following represents the most accurate definition of the term “persona”?

A. A final-version representation that includes colors and complex design decorations.	B. A representation of a type or pattern of users, based as far as possible on research or data about their goals, needs, behavior, and context.
C. A tool for evaluating a website’s server performance and technical response speed.	D. A design approach that prioritizes the comfort of the site’s creator and developers only.
2. What does a “wireframe” represent in the stages of building and planning websites?

A. A final visual design containing all colors and decorative fonts.	B. A back-end programming algorithm for organizing databases.
C. A preliminary layout for determining the placement and priority of elements, separate from visual design.	D. A tool for measuring the loading speed of Internet pages on smartphones.

Part II | Home Assignments

Essay Questions:

1. State the basic difference between designing a website based on the designer’s personal taste and relying on the “user-centered design (UCD)” approach.
.....
.....

2. What is meant by the terms “usability” and “accessibility” in the context of website design?

Multiple-Choice Questions: (choose the correct answer)

1. What is the term that refers to how easily and effectively a user can accomplish the required task within a website?
 - A. Accessibility.
 - B. Usability.
 - C. Visual contrast.
 - D. Structural proximity.
2. Which of the following accurately expresses the concept of “accessibility” in web design?
 - A. Designing the content and interface so that people with disabilities, including those with special needs, can use the site and interact with it.
 - B. Determining the placement and priority of elements using minimal visual detail.
 - C. Grouping related information close together on a single screen.
 - D. Unifying the font and color of headings across all pages of the site.

PERIOD 2

Week:

Date:/...../.....

Part I | Performance Tasks

Essay Questions:

1. Explain the principle of “contrast” from the CRAP principles, and show how a designer can apply it effectively to highlight the most important elements and make the priority of information clear to the user.
2. Explain in detail how the principle of “proximity” achieves the organization of information so the user perceives it as a single group at first glance.

Multiple-Choice Questions: (choose the correct answer)

1. Which of the following procedures represents a practical application of the principle of “repetition” from the CRAP principles in designing website pages?
 - A. Making headings huge and randomly changing their colors on every page.
 - B. Unifying the color, size, and font of headings on every page, and matching the design of bullet-point icons.
 - C. Distracting the user’s attention by scattering elements in far-apart corners of the screen.
 - D. Hiding the main navigation buttons inside complex sub-menus.

2. What is the basic rule on which the principle of “alignment” relies in designing Arabic-language user interfaces?
- A. Aligning the main text and elements strictly from left to right.
 - B. Distributing elements randomly without following a common baseline.
 - C. Aligning elements according to the language direction and the design grid, so that the main text direction is from right to left, using consistent rules.
 - D. Relying on complex geometric decoration to fill empty spaces.

Part II | Home Assignments

Essay Questions:

1. State the four principles of design known as the “CRAP” principles, and briefly explain the basic purpose of using them to organize the shape of web pages.
-
-
2. Explain how the design steps integrate, starting from picturing the “persona”, through the “wireframe”, to applying the “CRAP” principles, to reach a successful user-centered design.
-
-

Multiple-Choice Questions: (choose the correct answer)

1. What is the design principle, among the CRAP principles, concerned with grouping product images, names, and prices close together and leaving purposeful spacing between them?
- A. Contrast.
 - B. Repetition.
 - C. Alignment.
 - D. Proximity.
2. Which of the following represents an unsuitable choice or irrelevant piece of information when preparing a “persona” for the target user?
- A. The user’s purpose for using the website.
 - B. The website server’s performance and technical response speed.
 - C. The user’s age, profession, and context of use.
 - D. The challenges and difficulties the user faces.

PERIOD 3**Weekly Assessments**

Week:

Date:/...../.....

MODEL A

Essay Questions:

1. Discuss the importance of understanding the user and their context of use as a first step in good web design, and how this protects the design from falling into the trap of relying on the designer’s personal taste.
-
-

2. Explain the technical and design purpose of using a “wireframe”, showing why choosing colors, fonts, and decorations is postponed to later stages.
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.....
3. List the four principles of design (CRAP), stating the functional purpose of each principle in making the page easier to understand and improving the user’s browsing experience.
.....
.....
4. Explain in detail the role of the “user-centered design (UCD)” approach in improving usability indicators and digital accessibility for people with special needs.
.....
.....

Multiple-Choice Questions: (choose the correct answer)

1. What should good web design begin with?
 - A. Understanding the user, their needs, and their context of use, not just the designer’s taste.
 - B. Choosing luxurious colors and visual decorations first.
 - C. Writing back-end program code and databases.
 - D. Increasing hosting server speed only.
2. What is the correct description of a “wireframe”?
 - A. A preliminary layout for determining the placement and priority of elements, with minimal visual detail, before the final design.
 - B. A final design combined with high-resolution graphics and images.
 - C. A program file for executing electronic-payment operations.
 - D. A tool for testing mobile Internet browsing speed.
3. What is the purpose of applying the “contrast” principle in designing web pages?
 - A. Making headings clear and immediately highlighting important elements.
 - B. Unifying the font and color of headings across all pages.
 - C. Grouping related elements close together.
 - D. Aligning texts according to a common baseline.
4. Which of the following represents the accurate concept of user-centered design?
 - A. An approach that makes understanding users, their needs, and their context of use the basis for design decisions.
 - B. Giving priority to the comfort of the site’s creator and developers.
 - C. Relying absolutely on the individual designer’s taste.
 - D. Skipping the stage of evaluating and reviewing the site with actual users.

MODEL B**Essay Questions:**

1. Explain how building a “persona” based on real research or data helps the team discuss the needs of a particular type of user accurately.
.....
.....
2. Explain in detail how applying the “contrast” principle helps highlight important headings and buttons, while keeping readability in mind and not relying on color alone.
.....
.....
3. Show with examples how a web designer applies the “alignment” principle to create a consistent visual system on the screen according to the design grid and language direction.
.....
.....
4. Explain the complementary relationship between the user-centered design (UCD) life cycle and testing solutions with users to iteratively improve interfaces.
.....
.....

Multiple-Choice Questions: (choose the correct answer)

1. What is the basic benefit of using a “persona” in web projects?

A. Helping the team discuss the needs of a type of user instead of relying on personal assumptions.	B. Increasing the number of site visitors through search engines.
C. Reducing the size of image and media files on the page.	D. Speeding up the loading of web pages on smart devices.
2. Which of the four design principles is concerned with aligning the positions of elements to a common baseline to give the screen order?

A. Contrast.	B. Repetition.
C. Alignment.	D. Proximity.
3. What does the principle of “repetition” do in designing website pages?

A. It repeats certain characteristics such as color, size, and font to achieve consistency and help the user navigate without confusion.	B. It groups related elements together in one place.
C. It highlights important elements by creating clear visual differences.	D. It determines the placement and priority of elements separately from visual design.

4. What is the goal of studying “usability” and “accessibility”?

- A. Measuring the technical server’s performance speed.
- B. Improving the ease and effectiveness of accomplishing the task and enabling people with disabilities to use the site effectively.
- C. Reducing the cost of purchasing domains and hosting.
- D. Designing the program code for back-end languages.

MODEL C

Essay Questions:

1. Show how a carefully designed wireframe can help stakeholders and the technical team reach agreement on the distribution and placement of information, separate from visual appearance.

2. Compare the principle of “repetition” with the principle of “proximity” in terms of the visual effect each has on organizing the content of web pages.

3. Explain how ignoring “accessibility” standards deprives certain groups of users (such as those with low vision or disabilities) of effective interaction with a website’s services.

4. Discuss in detail the sequential steps of the user-centered design process, starting from studying needs, through organizing information, to evaluation and testing.

Multiple-Choice Questions: (choose the correct answer)

1. What is the design principle responsible for grouping related information (such as product images and their prices) close together?

- A. Proximity.
- B. Contrast.
- C. Repetition.
- D. Alignment.

2. What is meant by the concept of “user-centered design (UCD)”?

- A. A design approach that makes understanding users, their needs, and their context of use the basis for design decisions.
- B. An advanced method for programming databases.
- C. Designing interfaces that rely entirely on the developer’s personal intuition.
- D. A tool for evaluating network and Internet-connection errors.

3. What is the main reason for postponing the choice of colors, fonts, and decorations until after finishing the “wireframe”?
- A. To prevent discussions from drifting toward personal preferences about appearance, and to make it easier for stakeholders to agree on the information structure.
 - B. To increase the speed of graphics processing in the user’s browser.
 - C. To reduce the site’s storage space on the server.
 - D. To avoid using modern programming languages.
4. Which of the following represents the correct application of the “alignment” principle in Arabic texts?
- A. Making the main text direction from right to left, following consistent rules.
 - B. Making the text direction strictly from left to right.
 - C. Distributing words randomly in the corners of the screen.
 - D. Hiding texts inside complex drop-down menus.

UNIT 4 | WEB DESIGN AND MEDIA

LESSON

3

Methods for Evaluating Websites

PERIOD 1

Week:

Date:/...../.....

Part I | Performance Tasks

Essay Questions:

1. What is the basic difference between qualitative evaluation and quantitative evaluation in terms of focus and goal for each, in evaluating websites?
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2. Explain what is meant by “usability testing” as one of the qualitative evaluation tools, and how it contributes to improving the user interface.
.....
.....

Multiple-Choice Questions: (choose the correct answer)

1. Which of the following methods is a qualitative evaluation tool for understanding the user experience and the reasons behind behavior?

A. Usability testing.	B. Web analytics.
C. A/B testing.	D. Conversion rate (CVR).
2. What is the evaluation term that focuses on measuring with numbers and using numerical data and clearly defined metrics?

A. Qualitative evaluation.	B. Heuristic evaluation.
C. Quantitative evaluation.	D. Direct interviews.

Part II | Home Assignments

Essay Questions:

1. Briefly explain the meaning of the concept of “web analytics”, and how it is considered one of the quantitative evaluation tools.
.....
.....
2. State two main indicators in web analytics, stating the purpose of each.
.....
.....

Multiple-Choice Questions: (choose the correct answer)

1. What does the indicator (PV) represent in web analytics?
 - A. The proportion of users who achieved a specific goal.
 - B. The total number of times a page has been viewed.
 - C. The proportion of users who left after the first page.
 - D. The site's overall bounce rate.
2. What is the indicator that measures the proportion of users or sessions that achieved a targeted action such as registration or purchase?
 - A. Page views (PV).
 - B. Bounce rate.
 - C. Conversion rate (CVR).
 - D. Heuristic evaluation.

PERIOD 2

Week:

Date:/...../.....

Part I | Performance Tasks**Essay Questions:**

1. Define "heuristic evaluation", and by whom it is carried out. What is its relationship with user-interface rules?
.....
.....
2. Give three examples of the heuristic rules used in evaluating a user interface, and briefly explain one of them.
.....
.....

Multiple-Choice Questions: (choose the correct answer)

1. What is meant by the term "heuristic evaluation"?
 - A. Measuring behavior using numbers and numerical data.
 - B. A review carried out by experts on the user interface based on rules of thumb.
 - C. Testing two options to see which achieves better results.
 - D. Counting the total number of page visits.
2. Which of the following represents one of the basic heuristic rules for a user interface?
 - A. Bounce rate.
 - B. Total page views (PV).
 - C. Error prevention.
 - D. Conversion rate (CVR).

Part II | Home Assignments**Essay Questions:**

1. Why is combining qualitative evaluation and quantitative evaluation more useful when evaluating a website compared with relying on only one of them?
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.....

2. Briefly explain how “A/B testing” can be used to measure user behavior and design results.

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Multiple-Choice Questions: (choose the correct answer)

- Which of the following indicators expresses the proportion of sessions or users who left the site after viewing only the first page?
 - Page views (PV).
 - Conversion rate (CVR).
 - Bounce rate.
 - Heuristic evaluation.
- When web analytics shows a large increase in the number of views (PV) alongside a noticeable decrease in the conversion rate (CVR), this indicates:
 - All visitors completed their purchases successfully.
 - Many users reach the page but do not complete the target action.
 - There are no final visits to the site at all.
 - The design is completely successful without needing any qualitative evaluation.

PERIOD 3

Weekly Assessments

Week:

Date:/...../.....

MODEL A

Essay Questions:

- Compare qualitative evaluation with quantitative evaluation, showing the techniques used in each category.

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- Explain the page-views (PV) indicator and the conversion-rate (CVR) indicator, showing the importance of each and their limits of use.

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- Discuss “heuristic evaluation”, stating its definition, who carries it out, and the most important heuristic rules used in examining the user interface.

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- Explain why a quantitative indicator alone is not enough to explain why visitors leave the site, and how qualitative methods help close this gap.

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Multiple-Choice Questions: (choose the correct answer)

- Which of the two main categories focuses on the detailed understanding of the user experience and the reasons behind behavior?
 - Qualitative evaluation.
 - Quantitative evaluation.
 - Web analytics.
 - A/B testing.
- What does the page-views (PV) indicator measure in web analytics?
 - The proportion of users who left the site.
 - The total number of times a page has been viewed.
 - The proportion of completed conversions only.
 - The exact number of unique visitors.
- What is the term used for experts reviewing a user interface based on rules of thumb?
 - Heuristic evaluation.
 - Usability testing.
 - Bounce rate.
 - Quantitative evaluation.
- Which of the following represents the correct description of usability testing?
 - Measuring behavior using numbers and numerical data.
 - Observing actual operations to identify points that are hard to use.
 - Comparing two options to check which is better.
 - Analyzing web data automatically.

MODEL B**Essay Questions:**

- Explain how “usability testing” is carried out, and what its basic goals are in studying user behavior.

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- Discuss web analytics as a quantitative evaluation tool, explaining the variety of methods for calculating indicators and the importance of checking the definition of the tool used.

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- State the three heuristic rules mentioned in the lesson for the user interface, and explain the concepts of “visibility of system status” and “consistency”.

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- Explain how a website’s team can combine numerical (quantitative) evidence with user (qualitative) opinions to reach a comprehensive and accurate evaluation.

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Multiple-Choice Questions: (choose the correct answer)

1. What is the technique that begins by observing actual operations of using the site through observation and interviews?
 - A. A/B testing.
 - B. Usability testing.
 - C. Heuristic evaluation.
 - D. Web analytics.
2. What is the technique that compares two options or two designs to check which leads to better results?
 - A. Web analytics.
 - B. Heuristic evaluation.
 - C. A/B testing.
 - D. Usability testing.
3. What does the indicator (PV) represent in web-analytics tools?
 - A. The conversion rate for goals.
 - B. The total number of times a page has been viewed.
 - C. The site's bounce rate.
 - D. Experts' heuristic rules.
4. What is the term that expresses the proportion of sessions or users who left the site after viewing only the first page?
 - A. Conversion rate.
 - B. Page views.
 - C. Bounce rate.
 - D. Heuristic evaluation.

MODEL C**Essay Questions:**

1. Explain the role of direct interviews in qualitative evaluation, and how they help listen to users' opinions, needs, and dissatisfaction.
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2. Explain the idea of "A/B testing", and how it helps compare two options to check which leads to better results for the site.
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3. Explain in detail what is meant by "bounce rate", and why it is considered an indicator that needs interpretation according to the definition of the analytics tool used.
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.....
4. Discuss the great importance of combining "how much" data with "why" qualitative data, through a practical example of evaluating an online store.
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.....

Multiple-Choice Questions: (choose the correct answer)

1. Which of the following tools is used to listen to users' direct opinions to understand their needs and dissatisfaction?
 - A. Interviews.
 - B. Web analytics.
 - C. A/B testing.
 - D. Bounce rate.
2. What is the method or approach that measures behavior and results using numerical data and clearly defined metrics?
 - A. Qualitative evaluation.
 - B. Quantitative evaluation.
 - C. Heuristic evaluation.
 - D. Usability testing.
3. What are the platforms or tools that collect, measure, and analyze site-usage data to understand reach and engagement?
 - A. Web analytics.
 - B. Usability testing.
 - C. Direct interviews.
 - D. Heuristic evaluation.
4. Which of the following heuristic rules means making sure the design reduces the likelihood of users making mistakes?
 - A. Visibility of system status.
 - B. Consistency.
 - C. Error prevention.
 - D. Conversion rate.

UNIT 4 | WEB DESIGN AND MEDIA

LESSON

4

The Iterative Improvement Process for Websites

PERIOD 1

Week:

Date:/...../.....

Part I | Performance Tasks

Essay Questions:

1. Explain why improving a website is considered a repeated process rather than a task done once, and what role the PDCA cycle plays in organizing this improvement.
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2. Explain the steps of the continuous improvement cycle (PDCA), giving a simplified description of each of its four steps.
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Multiple-Choice Questions: (choose the correct answer)

1. What is the term describing a method that continuously advances improvement by repeating the four steps (Plan, Do, Check, Act)?

A. Design thinking.	B. The PDCA cycle.
C. Continuous improvement.	D. A/B testing.
2. In the "Plan" step within the PDCA cycle, what should the team do?

A. Implement changes to the site directly.	B. Measure results and compare them to the baseline.
C. Set a clear goal and create a plan for a suitable change.	D. Adopt what worked and adjust the plan.

Part II | Home Assignments

Essay Questions:

1. State the basic idea behind using the PDCA cycle, design thinking, and A/B testing in developing digital products, instead of assuming the first version is final.
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2. Explain with an example how the “Do” and “Check” steps are applied to improve the completion rate of a registration form on a website.

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Multiple-Choice Questions: (choose the correct answer)

1. What does the “Act” step in the PDCA cycle mean if the change achieved the required goal?
- A. Changing the plan completely and canceling the site. B. Adopting the change or expanding it.
- C. Starting the empathize stage over again. D. Measuring the bounce rate only, without any adjustment.
2. Which of the following represents the correct description of the concept of “continuous improvement”?
- A. Building the site once, perfectly, without any adjustments. B. Changing all the site’s elements at once.
- C. Improving a little each time, through repeated cycles. D. Relying only on developers’ opinions, without users.

PERIOD 2

Week:

Date:/...../.....

Part I | Performance Tasks

Essay Questions:

1. Define “design thinking”, and show how it helps understand the problem and solve it from the user’s perspective through its various stages.
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2. State the five stages of the design-thinking model, and briefly explain what is meant by the “empathize” and “define” stages.

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Multiple-Choice Questions: (choose the correct answer)

1. Which stage of design thinking involves recognizing the user’s experience and needs through observation, questions, and appropriate research?
- A. Define. B. Ideate.
- C. Empathize. D. Test.
2. Which of the following stages in design thinking means quickly creating a draft or prototype based on the generated ideas?
- A. Prototype. B. Test.
- C. Define. D. Empathize.

Part II | Home Assignments**Essay Questions:**

1. Explain how “A/B testing” allows website designers to compare two versions using real data, and how traffic is distributed between them.
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2. State two basic conditions explained in the lesson to ensure the success of “A/B testing” when trying to isolate the effect of a single specific change.
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Multiple-Choice Questions: (choose the correct answer)

1. What is the precise definition of A/B testing?
 - A. An experiment that compares two versions, A and B, according to a predefined metric, and distributes visits between them randomly.
 - B. A method for creating a preliminary layout showing the plan of a web page.
 - C. A computing approach that uses the properties of quantum mechanics to process information.
 - D. A thinking method for analyzing the site's numerical data only.
2. If, in an A/B test, you want to isolate the effect of changing one element (such as the color of a purchase button), what should you do?
 - A. Change all the elements of the page at the same time.
 - B. Make the difference between the two versions clear and limited to that one element only.
 - C. Rely on intuition without collecting enough data.
 - D. Avoid using any numerical metrics for measurement.

PERIOD 3**Weekly Assessments**

Week:

Date:/...../.....

MODEL A**Essay Questions:**

1. Explain in detail why improving a website is a repeated process rather than a task done once, showing the role of the PDCA cycle.
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2. Discuss the design-thinking model, showing that it is a flexible method with five stages that can be revisited rather than a strict linear path.
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3. Explain how “A/B testing” works in comparing two versions, and state the importance of defining the metric, the analysis plan, and a sufficient sample.

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4. Apply the PDCA cycle to a practical example of raising a registration-form completion rate from 40% to 50%, explaining the four steps.

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Multiple-Choice Questions: (choose the correct answer)

- What is the term for the improvement method that repeats the four steps Plan – Do – Check – Act?

A. Design thinking.	B. The PDCA cycle.
C. A/B testing.	D. The prototype.
- What is the term for the way of thinking used to discover problems and generate solutions from the user’s perspective through five steps?

A. The PDCA cycle.	B. A/B testing.
C. Design thinking.	D. Continuous improvement.
- What is the term for the method in which two versions are prepared and compared using data to see which is more effective?

A. The PDCA cycle.	B. Design thinking.
C. A/B testing.	D. Empathize.
- Which of the following best describes the “Test” stage in design thinking?

A. Understanding the user’s needs through observation and questions.	B. Precisely clarifying the problem to be solved.
C. Generating many ideas to solve the problem.	D. Having users try the prototype and collecting their feedback.

MODEL B

Essay Questions:

1. Explain how user feedback and usage data help identify what is working and what needs improvement on websites.

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2. Explain the steps of the PDCA cycle in detail, stating the description of each step (Plan, Do, Check, Act).

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3. State the five stages of design thinking, and briefly explain the “ideate” and “prototype” stages.

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4. Discuss the importance of making the difference between version A and version B clear and specific in A/B testing, and why changing several elements together measures the effect of the whole bundle rather than a single element.

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Multiple-Choice Questions: (choose the correct answer)

1. From the following options, choose what corresponds to the “Check” step in the PDCA cycle:

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| A. Set a goal of reducing the bounce rate by 10%. | B. Change the design of the homepage according to the plan. |
| C. Measure the bounce rate after the change and compare it to the baseline. | D. Adopt the change or expand its scope. |

2. From the following options, choose what is NOT a suitable way to conduct an A/B test:

- | | |
|---|---|
| A. Preparing two versions and comparing them using real data. | B. Changing several elements at the same time when wanting to isolate the effect of one element. |
| C. Making decisions only after collecting a sufficient amount of data. | D. Showing each version to actual users. |

3. What is the most suitable description of the concept of the website-improvement process?

- | | |
|--|--|
| A. If a website is built at first, there is no need for any improvement at all. | B. It is enough to improve the visual design of the package just once. |
| C. It is important to improve gradually through repeated evaluation and continuous improvement. | D. All improvements should be done using A/B testing alone, without planning. |

4. Which of the following best describes “design thinking”?

- | | |
|---|--|
| A. A method that uses purely numerical data to analyze server usage. | B. A way of thinking that leads to discovering problems and generating solutions from the user’s perspective. |
| C. A method in which experts identify interface problems based on purely abstract rules. | D. A method for creating a programming code diagram in JavaScript. |

MODEL C

Essay Questions:

1. Explain how iterative improvement helps the team learn from results instead of assuming the first version will be perfect.

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2. Discuss how improvement in design thinking moves back and forth (between empathize, define, and test) rather than in one strict direction.

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3. Show how a successful A/B test is designed for a website suffering from a low click-through rate on the purchase button, specifying the metric and how traffic is distributed.

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4. State the basic differences between the PDCA cycle, design thinking, and A/B testing in terms of purpose and method of implementation.

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Multiple-Choice Questions: (choose the correct answer)

1. A flexible way of understanding the problem from the user's perspective, trying solutions, and then improving them through five stages, is called:

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|--------------------|----------------------------|
| A. The PDCA cycle. | B. Design thinking. |
| C. A/B testing. | D. Continuous improvement. |

2. A quick, testable draft created based on the ideas generated in design thinking is:

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|--------------------|----------------------------|
| A. The prototype. | B. The classical bit. |
| C. Edge computing. | D. The verification cycle. |

3. An experiment that compares two versions, A and B, according to a predefined metric to determine the better option, is called:

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|--------------------|---------------------|
| A. The PDCA cycle. | B. Design thinking. |
| C. A/B testing. | D. The baseline. |

4. The cycle made up of (Plan, Do, Check, Act), widely used in business improvement and quality management, is:

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|---------------------|-----------------------|
| A. The PDCA cycle. | B. A/B testing. |
| C. Design thinking. | D. The empathy model. |